Lab – Mid Task

Development of a Comprehensive PDF Reader & Utility Application

Note: Copying code from another student will result in zero marks for both parties involved due to plagiarism.

Course:	Mobile Application Development
Instructor:	Muhammad Abrar Saddique
Submission Date:	Due Date: 26/10/2026
Total Marks:	100 (marks distribution provided at the end of this document)

Objective

The aim of this project is to develop a fully functional **PDF Reader application** using Flutter and SQLite. The application allows users to open, read, and manage PDF documents efficiently, while also offering a set of companion utility features such as **document scanning**, **PDF-to-image conversion**, and **file organization**. The app should provide a smooth reading experience, persistent data storage, and customization capabilities.

Project Description

You are required to create a PDF reader application that not only handles the basic viewing of PDF documents but also includes advanced functionality such as document scanning, PDF-to-image conversion, bookmarking, reading history, and data export options. The application should offer a user-friendly interface, robust local storage solutions, and customization capabilities.

Functional Requirements

1. Setup and Configuration

- Set up the Flutter environment and create a new Flutter project.
- Initialize a SQLite database to manage document metadata, bookmarks, reading history, and user preferences persistently.

2. User Interface

Design a simple and intuitive interface with the following views:

- **Library / All PDFs:** List of all PDF files imported or scanned by the user.

- **Favorites / Bookmarks:** Documents and pages marked as favorites or bookmarked by the user.
- **Recent / History:** Recently opened documents and last-read pages.
- **Scanner:** A dedicated view to scan documents using the device camera.

3. PDF Reading & Document Management

- **Open PDFs:** Users should be able to open PDF files from device storage and view them in-app.
- **Read & Navigate:** Support page navigation, zooming, scrolling, and jumping to a specific page.
- **Edit Metadata:** Users should be able to rename documents and modify document details (e.g., tags, description).
- **Delete Documents:** Implement functionality to remove documents from the library.
- **Bookmark Pages:** Allow users to bookmark specific pages and return to them later.
- **Mark as Favorite:** Allow users to mark documents as favorites for quick access.

4. Advanced / Utility Features

- **Document Scanner:** Use the device camera to scan physical documents, crop and enhance them, and save them as PDF files.
- **PDF to Image Converter:** Convert selected pages (or the entire PDF) into image files (JPG/PNG) and save or share them.
- **Customization Options:** Implement theme customization (e.g., light and dark mode, reading background color) and allow users to choose notification sounds.
- **Reading Progress Tracking:** Display a progress bar or percentage indicating how much of a document has been read, and allow users to resume from where they left off.
- **Export Functionality:** Enable users to export document metadata / reading history / bookmarks to CSV, PDF, and email.
- **Search:** Allow users to search within the library by file name, tags, or metadata.

5. Notifications

- Implement local notifications to remind users of reading goals, recently added documents, or scheduled reading sessions.

6. Testing

- Thoroughly test the application for functionality, usability, and responsiveness.
- Identify and fix any bugs or issues.

7. Submission on GitHub

- Submit the following source code (Lib and Android Folders + **pubspc.yaml**) in the "**Lab-Mid"** folder of the required GitHub repository that has already been shared.
- Apk, along with any necessary configuration files, and a **video demonstration** of the application **(Optional)**.

Recommended Tools

- **Flutter** for UI and application logic.
- **SQLite** for data management.
- Any additional Flutter libraries or plugins found necessary (e.g., syncfusion_flutter_pdfviewer, pdf, image_picker, camera, flutter_local_notifications, printing, share_plus).

Estimated Time Commitment

- **Total Time:** Approximately 42 hours.
- **Daily Commitment:** Aim to dedicate 1.5 to 2 hours per day.

Support and Resources

Access to Flutter's official documentation and various online tutorials. Weekly check-in sessions for project guidance and troubleshooting.

Marks Distribution for PDF Reader & Utility Application Project

#	Component	Description	Marks
1	Project Setup and Design	Includes setting up the Flutter environment and the initial design of the user interface.	10

#	Component	Description	Marks
2	Database Integration	Correct setup of SQLite and schema design for managing documents, bookmarks, and history.	10
3	PDF Reading & Document Management	Implementation of opening, reading, renaming, deleting, bookmarking, and marking documents as favorite.	20
4	Advanced / Utility Features	Scanner, PDF-to-image converter, customization options, progress tracking, and export functionalities.	20
5	Notification & Search Functionality	Local notifications and in-library search based on user-defined settings.	10
6	Scanner & Conversion Implementation	Correct implementation of document scanning and PDF-to-image conversion.	10
7	User Interface	Usability, aesthetics, and responsiveness of the application.	10
8	Documentation and Submission on GitHub	Quality of the code documentation, and Required Folders.	10
Total			**100**